

Automated Construction of Multi-Element MEAM Interatomic Potentials

A DFT-Informed Neural-Network Optimization Framework — Final Report

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PHYS 400 — Final Presentation





Every load-bearing part is sized by two numbers:

- $E = \sigma/\varepsilon$ — Young's modulus: stiffness (GPa).
- $\nu = -\varepsilon_{\text{lat}}/\varepsilon_{\text{axial}}$ — Poisson's ratio: transverse contraction.

For cubic crystals both follow from the two stiffness constants C_{11} and C_{12} — **the quantities this work targets from first principles.**

They set: beam deflection $\delta \propto 1/E$, buckling load $\propto E$, natural frequency $\propto \sqrt{E/\rho}$; bulk modulus $K = E/[3(1 - 2\nu)]$, stress concentration, toughness.

Stability (cubic, Born–Cauchy): $C_{11} > C_{12}$.

Representative structural applications

Application	Material	E (GPa)	ν
Aerospace fuselage	Al 7075	71–72	0.33
Turbine disk	Inconel 718	200	0.29
Orthopaedic im-plant	Ti 6Al–4V	110	0.34
Automotive struc-ture	Mg AZ91	45	0.35
Structural steel	Mild steel	200	0.30
MEMS resonator	Si (single xtal)	130	0.28

Same two constants span polymers ($\nu \rightarrow 0.4$) to steel — and fix whether a design survives load.



Interatomic potential. Closed-form approximation of the many-body potential energy $U(\{r_i\})$; forces obtained as $F_i = -\nabla_i U$.

Historical development:

- Pairwise (Lennard–Jones, Morse) — too crude for metals.
- Embedded Atom Method (1984) — good for FCC metals.
- **Modified EAM** (MEAM) [1]: adds angular terms — covers BCC, HCP and mixed bonding.

Density-functional theory (DFT).

First-principles electronic-structure method (Kohn–Sham equations [2]); here a plane-wave + pseudopotential basis in Quantum Espresso [3].

Quantities computed:

- Cohesive energy E_{coh}
- Equilibrium lattice a_0 , bulk modulus B
- Binary formation energies $E_{\text{form}}(i, j)$
- Single-crystal elastic constants C_{11}, C_{12}

DFT: high accuracy, limited scale ($\leq 10^3$ atoms).

MEAM/MD: empirical, large scale ($\geq 10^6$ atoms).

DFT provides the reference dataset against which the MEAM potential is calibrated.

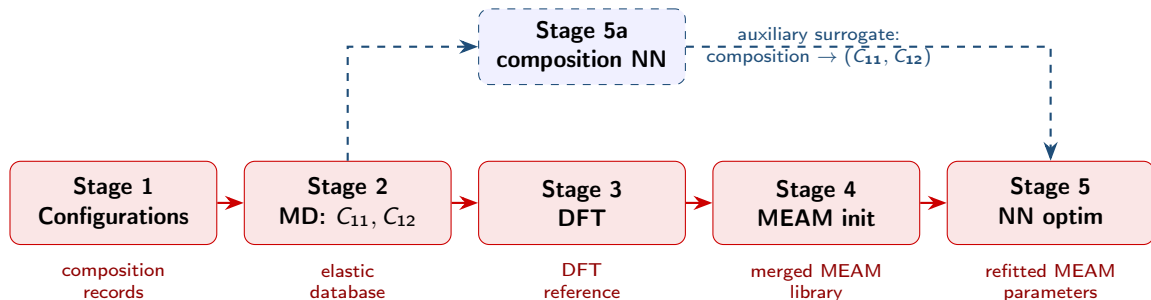


Published MEAM libraries cover disjoint subsets of elements; no single file spans the nine elements specified by Al 7075.

Library	Al	Co	Cr	Cu	Fe	Mg	Mn	Mo	Ni	Si	Ti	Zn
AlMgZn	✓					✓						✓
AlSiCuMgFe	✓			✓	✓	✓				✓		
CuMo				✓				✓				
FeMnNiTiCuCrCoAl	✓	✓	✓	✓	✓		✓		✓		✓	
Merged (this work)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Limitation of direct merging: cross-interaction parameters between elements drawn from distinct libraries are not defined; manual refitting does not scale to $\binom{12}{2} = 66$ binary pairs.

Pure machine-learning potentials [4–6]: require $10^4 - 10^5$ DFT configurations and forfeit the physical interpretability of MEAM.



The four published libraries are merged into a single 12-element file; cross-terms are refitted by inverting an NN surrogate of the MEAM $\rightarrow (C_{11}, C_{12})$ map under L-BFGS, with single-element parameters anchored to DFT references.



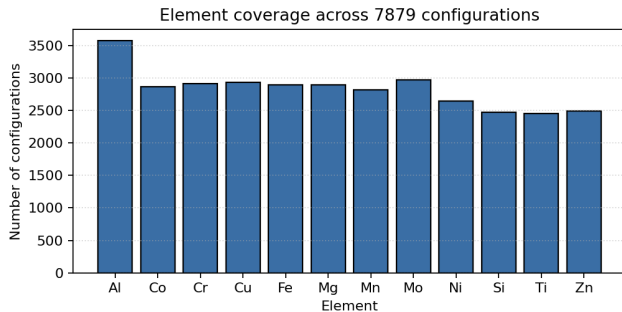
Sampling.

Uniform draws on the 12-element composition simplex (Al, Co, Cr, Cu, Fe, Mg, Mn, Mo, Ni, Si, Ti, Zn);
Al-weighted to oversample the aluminium-alloy region.

Corpus statistics.

~7900 configurations generated;

7815 retained after the physicality filter ($E > 0$, $0 < \nu < 0.48$, $C_{11} > C_{12}$).



Element-occurrence distribution over the retained corpus.



Implementation. [7]

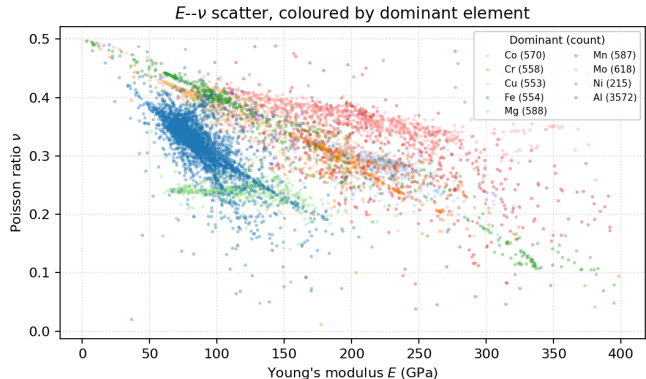
Custom LAMMPS build (MEAM species limit raised $5 \rightarrow 20$).

Per-configuration protocol:

- ~ 4 nm cubic supercell
- Atom-type reassignment to composition $\{c_i\}$
- Anisotropic energy minimisation
- Axial strain $\delta = 10^{-3}$ applied along $\pm x, \pm y, \pm z$
- Central-difference $\rightarrow C_{11}, C_{12}$

	Median	Std.	Range
E (GPa)	103.6	69.1	3 – 398
ν	0.326	0.063	0.01 – 0.50

$N = 7815$ retained configurations.



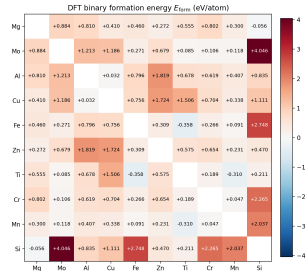
E - ν distribution, coloured by dominant element. Mg- and Zn-rich compositions cluster near $\nu \rightarrow 0.5$, motivating training on C_{11}, C_{12} rather than (E, ν) .

Stage 3–4: DFT References → MEAM Initialisation



Element	Lat.	a_0 (Å)	E_{coh}	B (GPa)
Al	FCC	4.047	3.611	77.5
Cu	FCC	3.639	3.857	138.5
Si	DIAMOND	5.471	5.130	89.3
Ti	HCP	2.915	6.284	114.5
Zn	HCP	2.766	0.964	68.4
Cr	BCC	2.848	3.964	213.1
Fe	BCC	2.806	6.834	420.5
Mg	HCP	3.177	1.483	47.8
Mn	BCC	2.803	3.728	225.7
Mo	BCC	3.169	10.353	280.8

Elemental references (E_{coh} in eV/atom; Birch–Murnaghan fits).



Binary formation energies $E_{\text{form}}(i, j)$ (eV/atom): negative = exothermic mixing, positive = limited solubility — these seed the MEAM cross-terms.

Stage 4 — single-element initialisation

Screening exponent from each DFT triple, $\alpha = a_0 \sqrt{9B\Omega/E_{\text{coh}}}$ (Ω : cell volume) → complete 12-element MEAM library (~ 70 ms, cached).
 α -Mn EOS did not converge → relaxed-BCC approximation.

Stage 5a: Composition $\rightarrow (E, \nu)$ Neural Surrogate



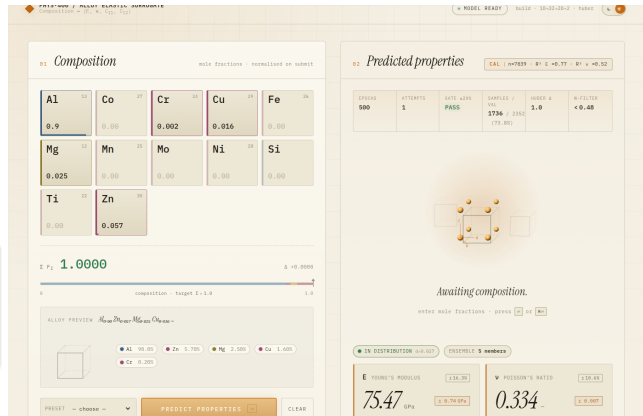
Architecture. $12 \rightarrow 32 \rightarrow 20 \rightarrow 2$
fully-connected MLP
with ReLU activation; Huber loss on $\hat{C}_{11}, \hat{C}_{12}$;
deep ensemble of five seeds [8] for
epistemic uncertainty.

Result

Held-out (30 % split): $R_E^2 = 0.77$, $R_\nu^2 = 0.52$.

AI 7075 preset: $\hat{E} = 75.5$ GPa, $\hat{\nu} = 0.334$
(published: 71–72 GPa, $\nu \approx 0.33$ — within $\sim 5\%$).

Distributed as a portable application image.



Interactive composition predictor with k -nearest-neighbour distance-based out-of-distribution flagging.



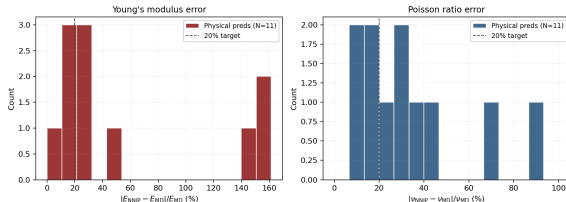
Closed-loop procedure:

- 1 NN surrogate learns (MEAM params) $\rightarrow (C_{11}, C_{12})$.
- 2 L-BFGS [9] minimises the Huber loss through the surrogate.
- 3 Optimised parameters are re-evaluated in LAMMPS; new samples are appended and the surrogate retrained.

Atomic checkpointing renders the procedure resumable.



NNIP Stage-5 verification on 30 held-out alloys | E within 20%: 4/30 | ν within 20%: 4/30 | unphysical NNIP outputs: 19/30



Result

Held-out alloys $\rightarrow$ pass stability filter 30 $\rightarrow$ 11
Within 20 % of MD on E / on ν 4 / 4
Median error on E / ν (physical) 23 % / 26 %

30 held-out $\rightarrow$ 11 stable $\rightarrow$ 4 within tolerance.
19 of 30 violate the Born–Cauchy criterion — the remaining gap motivates the Stage 5b extensions (next slide).

Surrogate Huber-loss decay from 1.0 to 0.16 over 10^4 steps.



Contributions

- MD elastic database, 7815 alloys
- DFT reference set, 10 elements
- Merged 12-element MEAM library
- Composition surrogate ($R_E^2 = 0.77$, deployed)
- Resumable Stage 5b framework

Current limitations

- 4/30 predictions within tolerance
- 19/30 violate Born–Cauchy
- Cross-term parameters saturate bounds

Identified causes

- Under-determined inverse problem (~ 200 parameters, ≤ 100 targets)
- Surrogate–LAMMPS residual mismatch
- Stability constraints absent from the loss

Planned extensions

- Restrict optimisation to single-element-anchored cross-terms ranked by Sobol sensitivity
- Active-learning expansion to ≥ 500 medoids
- Physics-informed loss with stability penalties
- SQS DFT references [10] at intermediate compositions



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